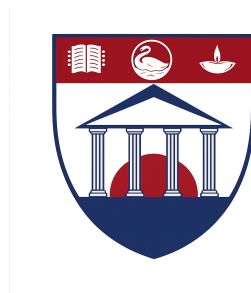


An Internship Report
Of
Datacom Software Development Job Simulation
(Forage Virtual Job Simulation)

Submitted
To
School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By
Roll Number of Student: 25SCS1003002955
Name of Student: ARUKANSH JAISWAL
Batch: 2025-2029

2026-27

CANDIDATE’S DECLARATION

I, ARUKANSH JAISWAL, do hereby declare that the internship report titled “Datacom Software Development Job Simulation” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:- _____

Student Name: ARUKANSH JAISWAL

Roll No.: 25SCS1003002955

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Forage and the software development team at Datacom for designing and making available the Software Development Job Simulation, which gave me a structured and realistic opportunity to experience the kind of work a software developer performs on the job. I am also thankful to my institute, IILM University, Greater Noida, and to the faculty of the School of Computer Science and Engineering for their continuous support, guidance and encouragement throughout the internship period.

I express my gratitude to my parents for their blessings.

Signature:- _____

Student Name: ARUKANSH JAISWAL

Roll No.: 25SCS1003002955

TABLE OF CONTENTS

Chapter No.	Chapter Name	Page No.
	Cover Page	----
1.	Candidate's Declaration	i
2.	Acknowledgement	ii
3.	Internship Completion Certificate	----
4.	Project Description 4.1 Introduction 4.2 Organization Profile 4.3 Problem Statement 4.4 Project Objectives 4.5 Scope of the Project 4.6 Technologies and Tools Used 4.7 System Architecture 4.8 Methodology 4.9 Expected Outcomes 4.10 Certificates of Completion and Communication Proof	1 1 2 2 3 3 4 4 5 6
5.	Bibliography / References	8

3. INTERNSHIP COMPLETION CERTIFICATE

DATACOM



Inspiring and
empowering
future professionals

Arukansh Jaiswal **Software Development Job Simulation**

Certificate of Completion
September 20th, 2026

Over the period of September 2026, Arukansh Jaiswal has completed practical tasks in:

Software review
Identifying root causes and fixing bugs

A stylized, handwritten signature in grey ink.

Tom Brunskill
Co-Founder of Forage

Enrolment Verification Code 6aaf9690875b6871a8b1eb04 | User Verification Code 6aaf94ca875b6871a8b1a38c | Issued by Forage

Certificate of Completion – Datacom Software Development Job Simulation, issued by
Forage on 20th September 2026.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the **Software Development Job Simulation** offered by **Datacom** on the **Forage** virtual work-experience platform, completed during September 2026 as part of the requirement of the Bachelor of Technology (Computer Science and Engineering) programme at IILM University, Greater Noida. A job simulation is a self-paced, industry-designed programme in which a student steps into the role of an employee of the partner organisation and completes the same category of tasks that the organisation's own team performs, guided by briefing material from the company and compared against model answers prepared by its engineers.

In this simulation the participant takes on the role of a software developer in Datacom's software development team and works on an existing web application belonging to a client. The programme is built around two practical tasks: reviewing the application and planning improvements for it, and then identifying the root causes of reported defects and fixing them. The primary aim of the internship was therefore to gain first-hand exposure to two activities that occupy a large part of a professional developer's working day but receive comparatively little attention in academic coursework – **structured software review** and **systematic root-cause analysis and debugging** of code that somebody else has written.

On completion of both tasks, a Certificate of Completion was issued by Forage, confirming that practical tasks in software review and in identifying root causes and fixing bugs had been completed. The certificate is attached in Chapter 3 of this report.

4.2 Organization Profile

Datacom Group Limited is one of Australasia's largest locally owned information technology services organisations. It was founded in 1965 in Christchurch, New Zealand, originally as Computer Bureau Limited, and has since grown into a group employing several thousand people across New Zealand, Australia and Asia. Datacom delivers IT management and consulting, cloud and data-centre services, custom software development, business process management and payroll services to large private-sector and government clients. Its software development practice builds and maintains business-critical web and enterprise applications, and the job simulation used in this internship is modelled directly on the review-and-fix work that this practice carries out for its customers.

Forage is an online platform that hosts free, self-paced virtual job simulations created in partnership with leading global employers. Each programme is authored by the partner company's own team, consists of a small number of realistic tasks supported by briefing videos and model answers, and concludes with a certificate of completion carrying enrolment and user verification codes. Forage programmes are designed to give students a risk-free, practical preview of professional work and to help them speak about that work with confidence in interviews.

4.3 Problem Statement

Undergraduate engineering students usually write software from scratch, for assignments whose requirements are already fixed and whose codebase they themselves created. Professional software development is very different. A developer typically joins a project in which the application already exists, the code was written by other people, the requirements keep evolving, and much of the work consists of reading unfamiliar code, judging its quality, and repairing defects reported by users or testers without breaking behaviour that currently works.

As a result, two skills that employers consistently look for remain underdeveloped at graduation: the ability to **review an existing application critically** and communicate clear, prioritised recommendations for improving it, and the ability to **trace a reported symptom back to its actual root cause** instead of applying a superficial patch to the place where the failure happens to surface. The problem addressed by this internship was to build and demonstrate these two capabilities in a realistic industry setting, by reviewing a live web application, proposing improvements to it, and then diagnosing and correcting defects in its code.

4.4 Project Objectives

- To understand how a software development team at an industry organisation such as Datacom approaches the review and improvement of an existing web application.
- To carry out a structured review of a web application, examining its functionality, user experience, code structure and maintainability.
- To document review findings clearly and to plan and prioritise a set of realistic improvements for the application.
- To read and comprehend a codebase written by somebody else and to reason about how its components interact.
- To reproduce reported defects reliably and to identify the underlying root cause of each defect rather than treating its visible symptom.

- To implement corrective fixes and verify that the intended behaviour is restored without introducing regressions elsewhere in the application.
- To apply critical thinking, systematic debugging and professional communication in a simulated workplace environment.
- To complete all the tasks required by the simulation and obtain the Certificate of Completion issued by Forage on behalf of Datacom.

4.5 Scope of the Project

The scope of this internship covers the two practical tasks that constitute the Datacom Software Development Job Simulation. The first task spans the review of an existing web application – assessing what the application does, how it is organised, where its weaknesses lie in terms of functionality, usability and maintainability – and the preparation of a written set of recommended improvements. The second task spans defect resolution: reproducing the reported issues, investigating the code to locate the responsible logic, determining the root cause, implementing a fix and confirming that the corrected behaviour is achieved.

The internship was conducted entirely in a virtual, self-paced simulation environment provided by Forage. It did not involve employment with Datacom, access to any live production system belonging to Datacom or its clients, or the handling of any real customer data, and no stipend was associated with it. The application worked on during the programme is a training artefact created by Datacom for educational purposes.

4.6 Technologies and Tools Used

- **Web application stack:** HTML, CSS and JavaScript – the front-end technologies in which the reviewed application is written.
- **Code editor / IDE:** Visual Studio Code, used for reading the codebase, navigating between files and implementing fixes.
- **Browser developer tools:** the console, elements inspector and network panel of a modern web browser, used to reproduce defects and observe runtime behaviour.
- **Debugging techniques:** breakpoint and step-through debugging, console logging, isolation of failing inputs and hypothesis-driven root-cause analysis.
- **Software review practices:** functional walkthrough, code-readability and maintainability assessment, and prioritisation of findings by impact and effort.
- **Documentation:** written review notes, improvement plans and defect explanations prepared in the format expected by the Datacom team.

- **Learning platform:** the Forage job-simulation platform, including task briefings, pre-recorded guidance from Datacom engineers and model answers used for self-comparison.

4.7 System Architecture

The work carried out during the simulation can be described as a layered workflow, in which each layer takes the output of the previous one as its input:

- **Application Layer:** the existing client web application supplied by Datacom, consisting of its user interface, client-side logic and supporting assets. This is the artefact under examination.
- **Review Layer:** a structured inspection of that application – what it is intended to do, how it actually behaves, how the code is organised, and which aspects of functionality, usability and maintainability need attention.
- **Planning Layer:** the consolidation of review findings into a prioritised improvement plan, communicated in writing to the team in the way a developer would report back to a project lead.
- **Diagnosis Layer:** reproduction of each reported defect, inspection of the relevant code paths and use of debugging tools to narrow the failure down to a single underlying cause.
- **Remediation Layer:** implementation of the corrective change at the point of the root cause, written so that it is consistent with the existing style and structure of the codebase.
- **Verification Layer:** re-testing of the corrected behaviour and of the surrounding functionality to confirm that the defect is resolved and that no regression has been introduced.
- **Assessment Layer:** submission of each task deliverable on the Forage platform and comparison against the model answer prepared by Datacom engineers, leading to the Certificate of Completion.

4.8 Methodology

The internship followed a task-wise methodology in which each task began with a briefing from the Datacom team, continued with independent hands-on work, and closed with a submitted deliverable that was compared against a model answer. The stages are summarised in Table 4.1 below.

Stage	Task / Module	Description of Work
1	Orientation and company briefing	Introduction to Datacom, to its software development team and to the client application to be worked on; understanding the role and expectations of a developer in the team.
2	Software review of the web application	Examination of the existing web application – its features, user experience, code structure and maintainability – and documentation of observations and potential problem areas.
3	Planning improvements	Consolidation of the review findings into a clear, prioritised set of recommended improvements, written in the form expected by the team.
4	Identifying root causes of bugs	Reproduction of the reported defects, inspection of the relevant code and use of browser developer tools and step-through debugging to trace each symptom back to its actual cause.
5	Fixing the bugs	Implementation of corrective changes at the root cause, consistent with the existing code style, followed by re-testing to confirm the fix and to check for regressions.
6	Submission and completion	Submission of the task deliverables on the Forage platform, comparison with the model answers provided by Datacom engineers, and issue of the Certificate of Completion.

Table 4.1: Task-wise methodology followed during the job simulation

Throughout the programme, each deliverable was first attempted independently and only then compared with the model answer supplied by the Datacom team. This comparison step was an important part of the methodology, because it showed not only whether a conclusion was correct but also how a professional developer would have structured, justified and communicated it. Note that this was an unpaid, self-paced virtual simulation and no stipend was associated with it.

4.9 Expected Outcomes

- Ability to review an unfamiliar web application methodically and to describe its strengths, weaknesses and areas for improvement in clear written form.
- Ability to prioritise improvement recommendations by their impact on the user and the effort required, rather than listing observations at random.
- Practical competence in reading and navigating a codebase written by somebody else and in reasoning about how its parts interact.

- A systematic, hypothesis-driven approach to debugging: reproduce the defect, isolate the failing component, identify the root cause, and only then change the code.
- Understanding of the difference between fixing a symptom and fixing a root cause, and of why the latter produces more maintainable software.
- Familiarity with browser developer tools and step-through debugging as everyday instruments of professional development work.
- Improved technical communication – explaining a defect, its cause and its fix in language a team or a client can follow.
- A completed job simulation and a Certificate of Completion issued by Forage on behalf of Datacom, confirming practical tasks in software review and in identifying root causes and fixing bugs.

4.10 Certificates of Completion and Communication Proof

The Certificate of Completion issued by Forage for the Datacom Software Development Job Simulation, bearing Enrolment Verification Code 6aaf9690875b6871a8b1eb04 and User Verification Code 6aaf94ca875b6871a8b1a38c, is attached in Chapter 3 of this report. The onboarding communication received at the start of the programme and the completion communication received on finishing it are attached below as further proof.

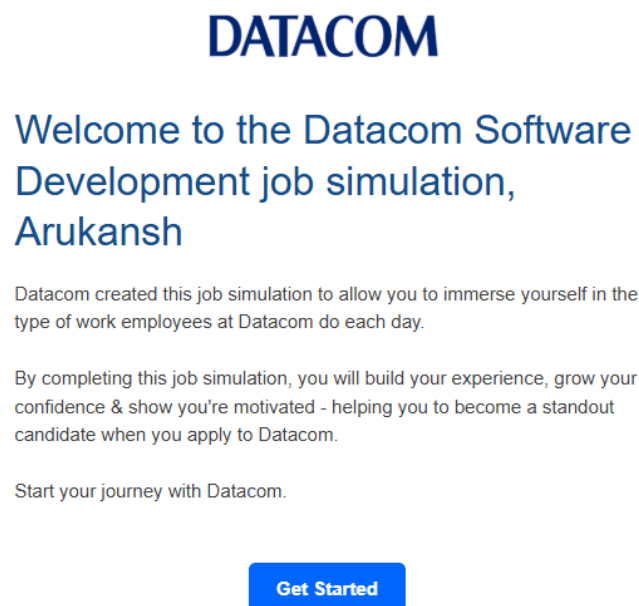


Figure 4.1: Welcome communication at the start of the Datacom Software Development job simulation



Figure 4.2: Completion confirmation received from Forage on finishing the job simulation

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